The Phenomenon: Why the Bilinear Form Works

ogbench-sota — phenomenon note, 2026-07-20
(companion: `PAPER/rescuing-fb/phenomenon.tex`, `PAPER/rescuing-fb/understanding-bilinear.tex`)

Factored-FB project

1 The question, and the facts any answer must reproduce

Swapping a monolithic MLP head $Q_\theta([s, a, g])$ for the normalized inner product $Q(s, a, g) = \phi(s, a)^\top \psi(g)/\sqrt{d}$ inside GC-IQL turns 0.71 into **0.90** on cube-single, transfers to GCIVL ($0.50 \to 0.79$) and HIQL ($0.15 \to 0.41$, once routed around the shared subgoal rep), and the same form recurs in essentially every method that has ever worked on task families: SF/USFA, FB, CRL, VIP, BVN, HILP/METRA, quasimetric nets. *Why?* Any explanation must reproduce all ten facts:

	Fact	Source
F1	Bilinear 0.904 vs MLP 0.713; the split is the *goal* axis (Δ_{GOAL}: bilinear $\in [-0.08, +0.08]$, MLP -16 to -24)	OOD grid
F2	The MLP is the *better* critic fully in-distribution (C1: 0.98 vs 0.87)	OOD grid
F3	The bilinear critic is *not* a better value function: counterfactual rank-AUC MLP 0.81 vs bilinear 0.39 (below chance); the MLP's signal is all in the action-independent baseline, the bilinear's baseline is anti-calibrated	cf-probe
F4	The bilinear critic's product is the *on-support action ordering*: frozen-critic reranking 0.91–0.96 (= its trained actor) vs MLP 0.57–0.59 (< its actor); FB's ordering is anti-informative (declines with N: $0.52 \to 0.17$)	rerank
F5	Action-dependence is $< 1\%$ of Q-variance for *every* critic (bilinear 0.46%, FB 0.21%, MLP 0.10%) — the loss barely sees the thing that decides success	variance probe
F6	Rank is a floor, not a dial: d=5 collapses (0.13), $d \in [28, 512]$ flat (0.86–0.93); at d=512 the "bottleneck" is as wide as the trunk and still beats the MLP by 16	rank sweep
F7	Eval-time goal smoothing cannot create the MLP's missing signal (monotone *decline*); if pooling is the mechanism it acts at training time	T1
F8	$\psi(g) \to \psi(s, g)$ destabilizes outright: 2/3 seeds diverge via a norm runaway on the state-conditioned goal leg; the stable seed lands at MLP level	bilinear_sg, T2
F9	Every *explicit* vector/feature-TD estimator of the same structure collapses (0.00–0.38); scalar TD through the same structure reaches 0.90	SF ladder
F10	The wins and failures are consumption-dependent and domain-dependent: cube flow 0.90 / DDPG+BC 0.51 (off-support peak, frac 0.639); antmaze DDPG+BC 0.837 (no peak, frac 0.003); scene loses with both actors (no peak — different failure)	T3, extraction

No single-clause story survives this table: "better values" dies on F3, "compression / regularization" dies on F6, "eval-time denoising" dies on F7, "data coverage" died on T3's dimension-corrected control. The explanation is a conjunction of four layers, each independently evidenced, and it is the conjunction — not any layer alone — that produces the cube result.

2 Layer 0 (representation): policy evaluation is genuinely bilinear

Proposition 1 (Task-linearity of evaluation)**.** *For any fixed policy π, the map $r \mapsto Q_r^\pi$ is linear: $Q_r^\pi(s, a) = \int M^\pi(s, a, \mathrm{d}s')\, r(s')$ with M^π the successor measure. Hence if a task family's rewards lie (ε-approximately) in a d-dimensional span, $r_g(s') \approx \beta(s')^\top w(g)$, then* exactly

$$Q_g^\pi(s, a) \;\approx\; \underbrace{\Big(\int M^\pi(s, a, \mathrm{d}s')\beta(s')\Big)}_{=:\,\phi^\pi(s,a)\ \textit{(the SF of } \beta\textit{)}}{}^\top w(g), \tag{1}$$

i.e. the bilinear head is not an approximation ansatz *but the* canonical form *of policy evaluation restricted to a low-rank reward family. The goal family of cube-single — indicator balls $r_g =$*

$\mathbf{1}[\|c(s) - g\| \le \varepsilon]$ *with g on a smooth 3-manifold — has a Gram operator with fast spectral decay (a smooth kernel on a 3-dim index set), so small d suffices and the required d is set by the task family's intrinsic complexity, not by state dimension.*

This single proposition already fixes three facts. **F6**: d must exceed the ε-rank of the reward family (the d=5 cliff = cannot span; the flat band = excess rank is unused because the target itself is low-rank — consistent with the empirical near-low-rankness of control Q-matrices, Yang et al. 2020; Shah et al. 2020). **F1**: for every goal, $Q(\cdot, \cdot, g) \in \text{span}\{\phi_1, \ldots, \phi_d\}$ — a *fixed* d-dimensional function space fitted against *all* training goals — so a novel evaluation goal requires extrapolating only the small map $\psi : \mathbb{R}^3 \to \mathbb{R}^d$, never the value function over (s, a); the MLP, nonlinear in g, must extrapolate the value itself and pays 16–24 points. **F9** (half): the structure being right says nothing about how to estimate it — see Layer 2.

Remark 1 (Value equivalence: the bilinear head spends its capacity exactly on the value span). The value-equivalence principle (Grimm et al. 2020, 2021) says a model/representation need only be correct on the span of value functions one will ever evaluate. Equation (1) is the value-equivalent compression of the environment for the task family: ϕ retains precisely the d occupancy moments the family's values need, and nothing else. The expectation-model line (Wan et al. 2019, 2021) supplies the closure property that makes TD work here: when values are linear in features, the Bellman backup needs only the *expected next feature* — a linear-in-ϕ value family is closed under expectation backups, so scalar TD on the head is a well-posed operator on the span. A monolithic MLP has no such closed span; each goal's backup is free to bend the function elsewhere.

Remark 2 (The caveat that produces the residual). The *optimal* value is not linear in r (the max breaks linearity), and $-V^*$ of goal-reaching is a quasimetric that symmetric inner products cannot represent with low distortion (QRL, IQE, MRN). So Proposition 1 can only hold approximately for the trained (expectile-improved) object, and the approximation error must land somewhere. Our account: the in-sample expectile keeps the trained object near the behavior-supported evaluation envelope where linearity is good, and the irreducible asymmetric residual is pushed *off-support* — observed as the off-support peak (**F10**: cube frac 0.639), whose location is a property of the domain's value geometry (antmaze 0.003), and as the scene failure, where the task family is combinatorial (multi-object) and a single rank-d product of one state tower and one goal tower is the wrong span. This is the one honest boundary of the explanation, and it is *predictive*, not hand-wavy: see Prediction 2.

Why the same form recurs across a decade of methods. Every historically successful task-family method is a rank factorization of the same evaluation functional, differing only in estimator:

Method	factorized object	estimator	weighting
SF/USFA	$Q^\pi_w = \psi^\pi(s,a)^\top w$	feature-TD + regression	task
FB	$M^{\pi_z} \approx F^\top B \rho$	density (measure) TD	occupancy $\rho \otimes \rho$
CRL	$\log \frac{p^\pi(s^+\|s,a)}{p(s^+)} \approx \phi^\top \psi$	InfoNCE on positives $\sim M$	trajectory-future
VIP	first-passage dual value $-\|\varphi(s) - \varphi(g)\|$	Fenchel-dual time-contrastive	goal-anchored
BVN	$Q(s,a,g) = f(s,a)^\top \phi(s,g)$	scalar TD (online, HER)	hindsight
GC-IQL bilinear (ours)	$Q = \phi(s,a)^\top \psi(g)/\sqrt{d}$	scalar expectile TD	hindsight mix

The form keeps working because the *object is genuinely bilinear* (Proposition 1); when one of these methods fails — FB on manipulation — the bridge program showed by exhaustive one-knob morphing that the failure is the *estimator column*, never the form (identical architecture: **87**

under scalar TD, **16** under the measure; the companion note carries that phenomenon). "Bilinear works" and "FB fails" are the same finding viewed from two sides: **structure ≠ estimator.** Our configuration is, in taxonomy terms, *successor features whose reward basis $\psi(g)$ is co-learned by scalar hindsight TD* — SFs without feature-TD.

3 Layer 1 (statistics): the shared basis pools a $<1\%$ signal across goals

Layer 0 says the form is right; it does not yet explain F3/F4 (ordering vs calibration) — why the bilinear critic *finds* an action ordering the MLP misses on the *same* data and loss.

Proposition 2 (Gradient coupling / cross-goal pooling)**.** *Decompose $Q(s,a,g) = U(s,g) + A(s,a,g)$ and the advantage into goal-common and goal-idiosyncratic parts, $A(s,a,g) = \bar{A}(s,a) + \epsilon(s,a,g)$ with $\mathbb{E}_g[\epsilon] = 0$. Under the bilinear head the TD gradient at $\phi(s,a)$ is*

$$\nabla_{\phi(s,a)}\mathcal{L} = \tfrac{1}{\sqrt{d}}\,\mathbb{E}_g\big[\delta(s,a,g)\,\psi(g)\big], \tag{2}$$

a linear *pooling of the TD errors of every relabelled goal along their ψ-directions: goal-common advantage errors accumulate coherently over the goal distribution while zero-mean idiosyncratic errors cancel at rate $O(N_g^{-1/2})$. The shared tower is a multi-task regression with a shared representation; its effective sample size for the goal-common component $\bar{A}$ is (state-visits × goals), not per-goal. A monolithic MLP receives only per-input gradients; with $\mathrm{Var}[A] < 10^{-2}\,\mathrm{Var}[Q]$ (F5), the variance-dominant baseline owns its capacity and optimization signal (and TD's implicit rank collapse pushes the same way), so its action signal is* absent*, not noisy — which is exactly what F7 measured. On cube the goal-common component is large by construction: post-grasp, "which action keeps the grasp while progressing" is nearly the same ordering problem for every transport goal (the phase diagnostic: the entire method spread is post-grasp, $P($*success* $|$ *secured*$)$ 0.98 *vs* 0.80*).*

This layer fixes **F3–F5, F7**, including the capacity-allocation *trade*: the rank-d head cannot simultaneously match an arbitrary baseline surface $U(s,g)$ (itself a rank-d-constrained matrix over states × goals) and the ordering; it spends its freedom on the pooled, goal-consistent structure and pays with calibration (anti-calibrated U, F3) — which is downstream-irrelevant, because extraction consumes orderings (Layer 3). The MLP makes the opposite trade: calibrated baseline (cf-AUC 0.81), no ordering.

Prediction 1 (T5, the discriminating control — pre-registered)**.** The pooling mechanism lives in the *linear* readout: only an inner product makes $\nabla_\phi\mathcal{L}$ a $\psi(g)$-weighted linear pool (2). A *late-fusion* control (identical two towers, MLP readout on $[\phi;\psi]$) re-mixes nonlinearly per input and breaks the coherent pooling. Prediction: late-fusion reranks near MLP level (≤ 0.65), far below bilinear (0.91–0.96), despite identical towers and parameters. A parameter-matched MLP (the 2× trunk confound control) stays at MLP level. If late-fusion instead reranks at bilinear level, Layer 1 is falsified and the explanation reduces to Layers 0/2 — this is the single most informative cheap experiment on the board.

4 Layer 2 (optimization): why scalar TD trains this structure and nothing else does

(a) Scale and grounding. Scalar TD pins units ($|Q| \leq (1-\gamma)^{-1}$, reward units); the $\sqrt{d}$ normalization keeps the product's scale $O(\|\phi\|\|\psi\|)$-stable — the same principle as bounded/structured

critic outputs (HL-Gauss, SimbaV2). Vector feature-TD and density-TD are self-referential: their fixed points are invariant to invertible linear reparameterizations, admit collapse (F=0-type) solutions unless anchored, and their anchors import density units with unbounded dynamic range — the measured runaways ($q \to 10^{13}$; SG-ISFB inflation) and the F9 ladder (0.00–0.38 across every explicit vector-TD estimator of the very same structure) are one defect. *Scalar TD through the structure* is the unique tested estimator that both grounds the scale and lets the structure emerge.

(b) The $\psi(g)$ bottleneck as global coupling (F8). With ψ a function of g alone, any local score inflation at (s, g) must move *globally shared* features and is corrected by the rest of the data; $\psi(s, g)$ opens a local (s, g) pocket that expectile TD's upward pressure (no output bound, no V-target on that leg) inflates without opposition — T2 located the divergence precisely there ($\mathbb{E}\|\psi\|^2$ 1083 $\to$ 2355 $\to$ 3789 monotone in severity, ϕ sane throughout). The same coupling is what lets goal-shift generalization be free (F1): the head cannot even *represent* a goal-idiosyncratic exception. Note the regime inversion vs online BVN (which prefers $f(s, a)^\top \phi(s, g)$): the offline expectile backup with no output bound is the standing suspect, testable by bounding the $\psi(s, g)$ leg (implementation note §7).

5 Layer 3 (consumption): the product is an ordering, and the consumer must be anchored

The trained bilinear critic's asset is the on-support ordering (F4); its one structured flaw is the off-support maximum (Remark 2). Extraction decides which of the two you consume: BC-anchored flow / selection-by-argmax consumes exactly the ordering and never visits the flaw (0.90–0.96); gradient ascent (DDPG+BC) walks distance 17.9 (whitened) off-manifold to the fabricated peak (0.51), while the MLP — no peak, no ordering — is indifferent (≈ 0.73 either way). Domains flip the sign because the peak's existence is domain value-geometry (F10). The corollary design rules, both verified: (i) on manipulation data the BC anchor is *mandatory* (the support argument in the companion note: the stable-vs-slipping boundary has ρ-mass 1.7×10^{-5} — no critic may be asked to rank stability counterfactuals); (ii) *the bilinear head must not train a representation the policy shares* (HIQL: routing the bilinear TD loss around ζ flips 14.5 $\to$ 41.3).

6 The explanation, assembled

Claim 1 (Why the bilinear form works)**.** $Q(s, a, g) = \phi(s, a)^\top \psi(g)/\sqrt{d}$ trained by scalar hindsight expectile TD works because four independently-evidenced layers align: **(0)** the target object — policy evaluation over a low-complexity task family — is *itself* bilinear (Prop. 1), so the head is the value-equivalent parameterization, rank cost is a floor set by the family (F6), and goal generalization is linear-span reuse, not extrapolation (F1); **(1)** the linear readout turns TD into a cross-goal gradient pool (Prop. 2) that accumulates the goal-common, $< 1\%$-variance action ordering the task actually needs, at the price of per-goal calibration it does not need (F3–F5, F7); **(2)** scalar TD is the unique tested estimator that grounds this structure's scale and reward-relevant subspace — vector/density TD on the identical structure collapses (F9) — and the g-only ψ leg globally couples the fit, closing the local-inflation pockets that destroy $\psi(s, g)$ (F8); **(3)** anchored/selection extraction consumes exactly the ordering the head produces and never visits the quasimetric residual it necessarily exiles off-support (F10). The historical record (SF, FB, CRL, VIP, BVN) is layer 0 recurring under different estimators; the FB-on-manipulation failure is a layer-2/weighting failure of one estimator, not of the form. Where any layer's precondition fails, the form fails exactly as predicted: rank below the family's span (d=5), a compositional family

outside one product's span (scene), gradient consumption of the residual (cube DDPG+BC), or a shared representation trained by the head (HIQL form a).

What makes this account hard to challenge is not rhetoric but coverage and falsifiability: every clause carries a completed experiment, the rival single-clause stories are each dead on a named fact, and the two genuinely open sub-questions have pre-registered discriminating experiments (Prediction 1; Prediction 2) whose either outcome *refines* the layer structure rather than overturning F1–F10, which are measurements.

Prediction 2 (Peak geometry — pre-registered)**.** If the off-support peak is the exiled quasimetric residual, its location should track embedding-norm growth off-manifold: $\|\phi(s,a)\|$ along rays $a_{\text{ref}} \to a_{\text{DDPG}}$ grows on cube and not on antmaze, and a norm-capped ϕ (spectral or hard cap) shrinks the cube peak fraction from 0.639 toward the antmaze 0.003 while leaving flow-actor success within noise. If the peak survives norm capping, the residual-relocation reading is wrong and the peak needs a new account — but the extraction rule (Layer 3) stands on its own measurements either way.

7 Final section: proposed experiments and implementations

Ordered by information-per-GPU-hour; each is specified concretely in `implementations.md` (this folder).

1. **T5 late-fusion + param-matched MLP** (Prediction 1). Two towers identical to bilinear; readout MLP($[\phi;\psi]$); rerank protocol at N=32 plus full training run; second arm: MLP with 2.06× width to close the parameter confound. Decides Layer 1.
2. **Norm-capped ϕ / peak-geometry probe** (Prediction 2). Recompute the T3 peak statistic along reference→DDPG rays vs $\|\phi\|$; then retrain cube DDPG+BC with a capped ϕ. Decides the residual-relocation reading and, if positive, yields a fix that makes gradient extraction safe.
3. **srank/effective-rank trajectories** of MLP vs bilinear ϕ during training — the implicit-under-parameterization half of Layer 1's "absent, not noisy"; cheap, uses existing checkpoints.
4. **T9: rank of the true advantage.** Estimate $\text{rank}_\varepsilon(A^*(s,\cdot,g))$ over the goal grid from counterfactual rollouts (atlas banks + simulator). Tests Layer 0's premise that the task family's ordering demands are genuinely low-rank; calibrates the d floor against ground truth.
5. **Bounded $\psi(s,g)$** (tanh output bound or V-target on the goal leg). Tests whether F8's divergence is the only obstruction to the BVN pairing offline — if it stabilizes at bilinear level, the online/offline inversion is an optimization artifact, and the paper's claim narrows cleanly to "g-only ψ is *sufficient* and *safe*, state-conditioning is fixable but unnecessary."
6. **Scene split-head**: per-factor goal towers $\sum_k \phi_k(s)^\top \psi_k(g_k)$ (Schaul-style factored family). Tests Remark 2's account of scene as a span failure of one product; a positive result extends Layer 0 to compositional families and likely recovers the scene loss.
7. **Cross-pollination (the bridge back to research question 2).** Layer 1 says the mechanism is *training-time* pooling through a shared basis under scalar TD; the companion note's RG-FB v1 (task-prior scalar grounding) is exactly "make the pooling distribution richer than Dirac goals." The two programs share one sentence: *the bilinear form is the right structure because evaluation is bilinear; scalar task-weighted TD is the right estimator because it is the only one that grounds, pools, and stays on support.*